A DICHOTOMY FOR DENSE LAMBDA-PARTITE ORDERED HYPERGRAPH COLLECTIONS WITH BOUNDED CLIQUE SIZES

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Allowed standard prerequisites (editorial).

- Ordered r-patterns, the pattern family of an ordered partition, P-cliques and their sizes, and the notion of a lambda-partite collection. Set up in the author's preliminaries (source lines 225-281) and in the author's definition at source lines 289-291. Fixes every object of the statement: the collection, its pattern family and the clique-size bounds.

Source reference. Original definition 2.1: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 289–289. Navigation pointer only: the counted proof refers to the author's definition of a partite collection; its content is declared as a named standard prerequisite and is not reproduced.

Source reference. Original lemma 3.1: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 359–359. Navigation

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pointer only: the counted proof invokes the author's interval-wise lemma; that statement is counted by another line's card LIT-comb-omr-interval-wise on the other staged copy of this paper, so it is a pointer here and is declared as a named standard prerequisite.

Source reference. Original theorem 1.2: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 199–199. Navigation pointer only: the counted proof refers to the author's upper-bound theorem for the sizes appearing in the alternatives; it is not reproduced.

Proposition 0.1. *Let r be a positive integer, and let $\lambda = (\lambda_1, \dots, \lambda_s)$ be an ordered partition of r into $s < r$ parts. For every ordered partition λ' of r into $s+1$ parts with $\lambda \succ \lambda'$, consider some positive integer $M_{\lambda'}$, and let $M = \sum_{\lambda'} M_{\lambda'}$ be the sum of all these positive integers. Let $\mathcal{H}$ be a finite collection of pairwise disjoint subsets $e \subseteq \mathbb{Z}$, each of size $|e| = r$, such that $\mathcal{H}$ is λ-partite. Then at least one of the following two statements holds:*

(a) *There is a sub-collection $\mathcal{H}^* \subseteq \mathcal{H}$ of size $|\mathcal{H}^*| \geq |\mathcal{H}|/(2M)$ such that $\mathcal{H}^*$ is interval-wise λ-partite.*
(b) *For some ordered partition λ' of r into $s+1$ parts with $\lambda \succ \lambda'$, there is a sub-collection $\mathcal{H}' \subseteq \mathcal{H}$ of size $|\mathcal{H}'| > M_{\lambda'}$ such that $\mathcal{H}'$ is λ'-partite.*

Proof of Proposition 0.1. Suppose that $\mathcal{H}$ does not have a subcollection $\mathcal{H}'$ as in (b). We need to show that then (a) holds. Let $x_0 < x_1 < \dots < x_s$ be as in Definition 2.1. For every $e \in \mathcal{H}$, we consider the intervals $e \cap (x_{i-1}, x_i)$ for $i = 1, \dots, s$. Our goal is to find a subcollection $\mathcal{H}^* \subseteq \mathcal{H}$ of size $|\mathcal{H}^*| \geq |\mathcal{H}|/(2M)$ such that for each $i = 1, \dots, s$, the intervals $\mathrm{conv}(e \cap (x_{i-1}, x_i))$ are pairwise disjoint for all $e \in \mathcal{H}^*$. Note that for any $i = 1, \dots, s$ and any $e \in \mathcal{H}$, the interval $\mathrm{conv}(e \cap (x_{i-1}, x_i))$ is a closed sub-interval of (x_{i-1}, x_i). Furthermore, the endpoints of all these sub-intervals are distinct (as all $e \in \mathcal{H}$ are pairwise disjoint subsets of $\mathbb{Z}$).

Let us consider an auxiliary graph G with vertex set $\mathcal{H}$, where for $e, e' \in \mathcal{H}$ we draw an edge between e and e' if $\mathrm{conv}(e \cap (x_{i-1}, x_i)) \cap \mathrm{conv}(e' \cap (x_{i-1}, x_i)) \neq \emptyset$ for some $i \in \{1, \dots, s\}$. Our goal is then to show that the graph G has an independent set of size at least $|\mathcal{H}|/(2M)$.

Note that for any two closed intervals $[a, b]$ and $[c, d]$ with $[a, b] \cap [c, d] \neq \emptyset$ we have $a \in [c, d]$ or $c \in [a, b]$. Hence we can orient the edges of G in such a way that for any $e, e' \in \mathcal{H}$ with an edge from e to e' there is an index $i \in \{1, \dots, s\}$ with $\min(e \cap (x_{i-1}, x_i)) \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$.

In order to show that the graph G contains an independent set of size at least $|\mathcal{H}|/(2M)$, it suffices to show that G has average degree at most $2M - 1$ (by a well-known result of Caro [3] and Wei [18]). Considering the orientation of the edges of G, it therefore suffices to show that for every $e \in \mathcal{H}$ there are at most $M - 1$ outgoing edges at e in the graph G (then there are at most $(M-1)|\mathcal{H}|$ total edges in G and hence the average degree is at most $2M - 2$).

So let us now fix some $e \in \mathcal{H}$. We need to show that there are at most $M - 1$ different $e' \in \mathcal{H} \setminus \{e\}$ such that $\min(e \cap (x_{i-1}, x_i)) \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$ for some index $i \in \{1, \dots, s\}$.

For every index $i \in \{1, \dots, s\}$ with $\lambda_i = 1$, we have $|e' \cap (x_{i-1}, x_i)| = \lambda_i = 1$ for every $e' \in \mathcal{H}$, and the singleton sets $e' \cap (x_{i-1}, x_i)$ are all distinct for all $e' \in \mathcal{H}$ (as all $e' \in \mathcal{H}$ are pairwise disjoint). Hence we cannot have $\min(e \cap (x_{i-1}, x_i)) \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$ for any $e' \in \mathcal{H} \setminus \{e\}$ when $\lambda_i = 1$.

We claim that for each $i \in \{1, \dots, s\}$ with $\lambda_i \geq 2$, there can be at most $\sum_{\lambda'} M_{\lambda'} - 1$ different $e' \in \mathcal{H} \setminus \{e\}$ with $\min(e \cap (x_{i-1}, x_i)) \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$, where the sum is over all ordered partitions λ' of r into $s+1$ parts that can be obtained from λ by dividing the i-th part λ_i into two pieces. This suffices to finish the proof, since

then the total number of $e' \in \mathcal{H} \setminus \{e\}$ with $\min(e \cap (x_{i-1}, x_i)) \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$ for some $i \in \{1, \dots, s\}$ is at most

$$\sum_{\substack{i \in \{1,\dots,s\} \\ \lambda_i \geq 2}} \left(\left(\sum_{\substack{\lambda' \text{obtained from } \lambda \text{ by} \\ \text{dividing } \lambda_i \text{into two parts}}} M_{\lambda'} \right) - 1 \right) = \sum_{\lambda'} M_{\lambda'} - |\{i \in \{1, \dots, s\} \mid \lambda_i \geq 2\}| \leq \sum_{\lambda'} M_{\lambda'} - 1 = M - 1,$$

where in the second and third term the sum is over all ordered partitions λ' of r into $s+1$ parts with $\lambda \succ \lambda'$ (note that each of these ordered partitions is obtained by dividing some part λ_i of λ with $\lambda_i \geq 2$ into two parts). Also note that for the inequality sign we have used that there must at least one index $i \in \{1, \dots, s\}$ with $\lambda_i \geq 2$, since $s < r$.

It remains to prove the above claim, so fix some index $i \in \{1, \dots, s\}$ with $\lambda_i \geq 2$. There are $\lambda_i - 1$ different ordered partitions λ' that can be obtained from λ by dividing λ_i into two parts, namely the ordered partitons $\mu^{(1)} = (\lambda_1, \dots, \lambda_{i-1}, 1, \lambda_i - 1, \lambda_{i+1}, \dots, \lambda_s)$, $\mu^{(2)} = (\lambda_1, \dots, \lambda_{i-1}, 2, \lambda_i - 2, \lambda_{i+1}, \dots, \lambda_s)$, $\dots$, $\mu^{(\lambda_i - 1)} = (\lambda_1, \dots, \lambda_{i-1}, \lambda_i - 1, 1, \lambda_{i+1}, \dots, \lambda_s)$. Defining $y = \min(e \cap (x_{i-1}, x_i)) \in \mathbb{Z}$, we need to show that there are at most $\sum_{j=1}^{\lambda_i - 1} M_{\mu^{(j)}} - 1$ different $e' \in \mathcal{H} \setminus \{e\}$ with $y \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$. Note that for every such $e' \in \mathcal{H} \setminus \{e\}$, we also have $y + 0.1 \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$, since $\mathrm{conv}(e' \cap (x_{i-1}, x_i))$ is a closed interval with endpoints in $\mathbb{Z} \setminus \{y\}$. We furthermore also have $y + 0.1 \in \mathrm{conv}(e \cap (x_{i-1}, x_i))$ (recalling that $y = \min(e \cap (x_{i-1}, x_i))$ and that $e \cap (x_{i-1}, x_i) \subseteq \mathbb{Z}$ is a set of size $\lambda_i \geq 2$), which in particular implies $x_{i-1} < y + 0.1 < x_i$. It now suffices to show that there are at most $\sum_{j=1}^{\lambda_i - 1} M_{\mu^{(j)}}$ different $e' \in \mathcal{H}$ with $y + 0.1 \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$. For every such e', we have $1 \leq |e' \cap (x_{i-1}, y + 0.1)| \leq \lambda_i - 1$ (recalling that $e' \subseteq \mathbb{Z}$ and $|e' \cap (x_{i-1}, x_i)| = \lambda_i$).

For each $j = 1, \dots, \lambda_i - 1$, there can be at most $M_{\mu^{(j)}}$ different $e' \in \mathcal{H}$ with $|e' \cap (x_{i-1}, y + 0.1)| = j$. Indeed, for each such $e' \in \mathcal{H}$ we have $|e' \cap (y + 0.1, x_i)| = |e' \cap (x_{i-1}, x_i)| - |e' \cap (x_{i-1}, y + 0.1)| = \lambda_i - j$. Hence, by considering $x_0, \dots, x_{i-1}, y + 0.1, x_i, \dots, x_s \in \mathbb{R} \setminus \mathbb{Z}$, we can observe that the collection $\mathcal{H}' \subseteq \mathcal{H}$ of all $e' \in \mathcal{H}$ with $|e' \cap (x_{i-1}, y + 0.1)| = j$ is $\mu^{(j)}$-partite. So by our assumption that (b) does not hold, there can be at most $M_{\mu^{(j)}}$ different such $e' \in \mathcal{H}$ (recall that $\mu^{(j)}$ is an ordered partition of r into $s+1$ parts with $\lambda \succ \mu^{(j)}$).

Thus, in total there can be at most $\sum_{j=1}^{\lambda_i - 1} M_{\mu^{(j)}}$ different $e' \in \mathcal{H}$ with $1 \leq |e' \cap (x_{i-1}, y + 0.1)| \leq \lambda_i - 1$, meaning that there are at most $\sum_{j=1}^{\lambda_i - 1} M_{\mu^{(j)}}$ different $e' \in \mathcal{H}$ with $y + 0.1 \in \mathrm{conv}(e' \cap (x_{i-1}, x_i))$. This finishes the proof of the proposition. □

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